Aga Khan Higher Secondary School Karimabad

Introducing Sync

SYNC is an AI-powered accessibility platform that uses computer vision and machine learning to recognize **Pakistani sign language gestures** and convert their meaning into understandable text and speech. Its primary objective is to make communication more accessible between sign-language users and people who do not understand sign language and contribute to our culture and arts further.



Team:

M Hasan

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Screenshots

```
optimizer: 'optimizer=auto' found, ignoring 'lr0=0.01' and determining best 'optimizer' and 'lr0' automatically...
optimizer: AdamW(lr=0.002, momentum=0.9) with parameter groups 126 weight(decay=0.0), 186 weight(decay=0.0005), 186 bias(decay=0.0)
Using 18392 train, 7847 val images for fraction=1.0 at imgsz=640
Using 2 dataloader workers
Logging results to /content/drive/MyDrive/sync_runs/hand_pose-2
Starting training for 30 epochs...

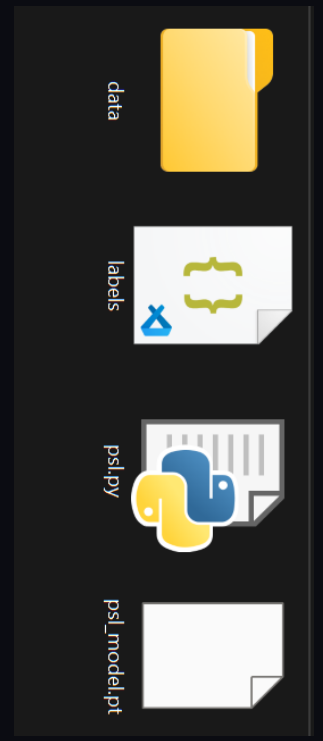
      Epoch    GPU_mem   box_loss  pose_loss  kobj_loss   cls_loss    l1_loss   rle_loss  Instances       Size
       1/30      2.96G      1.921      5.643     0.5527      2.563    0.03814      9.626         19        640: 100% ──────────── 1150/1150 2.8it/s 6:51
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-95)     Pose(P          R      mAP50  mAP50-95): 100% ──────────── 246/246 2.9it/s 1:26
                   all       7847       7847      0.921      0.771      0.926      0.605      0.867      0.467      0.515      0.319

      Epoch    GPU_mem   box_loss  pose_loss  kobj_loss   cls_loss    l1_loss   rle_loss  Instances       Size
       2/30      3.52G      1.471      3.933     0.3995      1.699    0.02868      5.268         16        640: 100% ──────────── 1150/1150 2.8it/s 6:50
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-95)     Pose(P          R      mAP50  mAP50-95): 100% ──────────── 246/246 2.9it/s 1:25
                   all       7847       7847      0.936      0.881      0.962      0.689      0.834      0.614      0.657      0.422

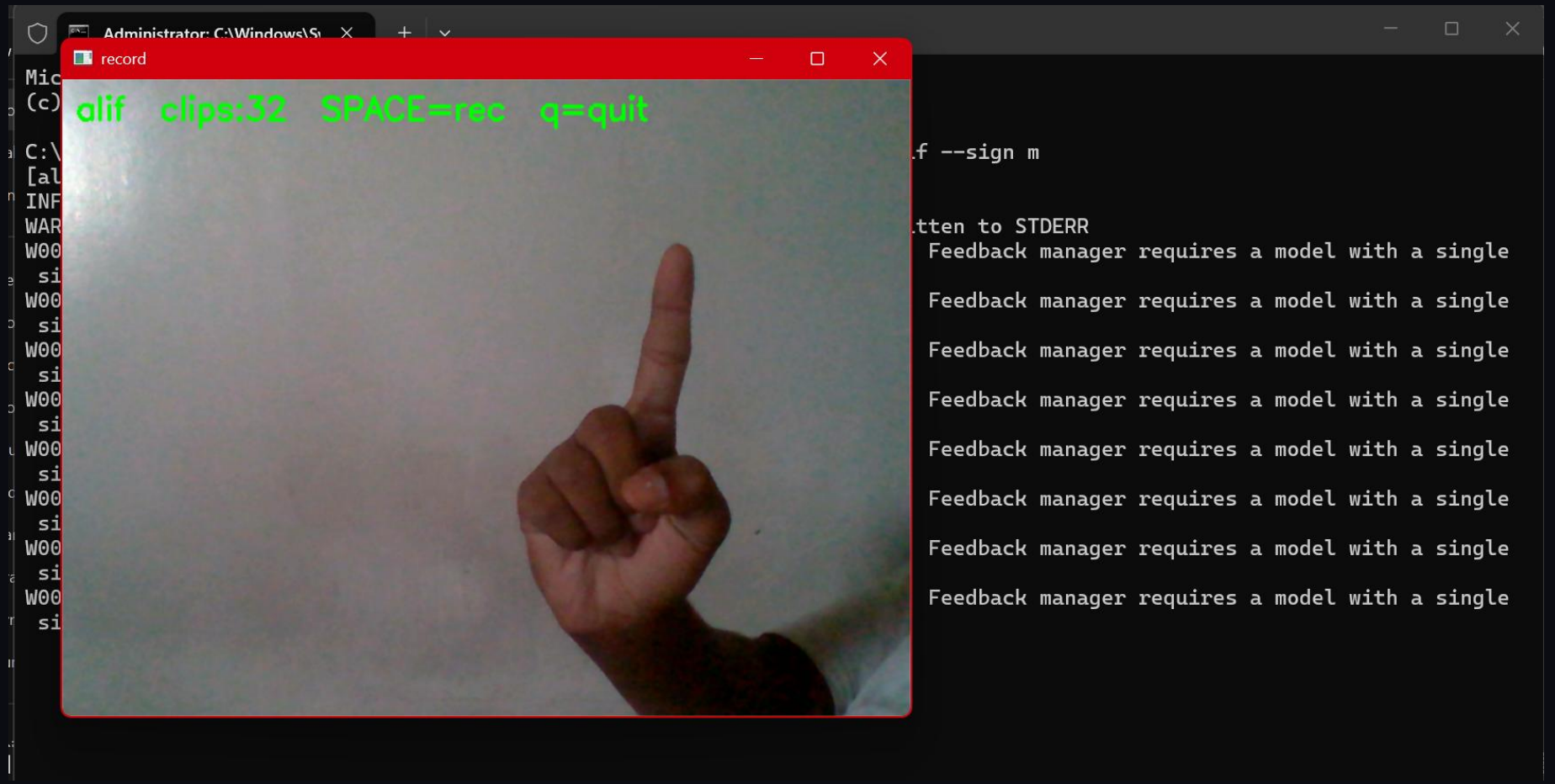
      Epoch    GPU_mem   box_loss  pose_loss  kobj_loss   cls_loss    l1_loss   rle_loss  Instances       Size
       3/30      3.52G      1.386      3.457     0.3327      1.513    0.02674      4.426         13        640: 100% ──────────── 1150/1150 2.9it/s 6:43
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-95)     Pose(P          R      mAP50  mAP50-95): 100% ──────────── 246/246 3.0it/s 1:22
                   all       7847       7847      0.932      0.906      0.968      0.731      0.863      0.662      0.695      0.477

      Epoch    GPU_mem   box_loss  pose_loss  kobj_loss   cls_loss    l1_loss   rle_loss  Instances       Size
       4/30      3.52G      1.289      3.145     0.2989      1.397    0.02461      3.875         14        640: 100% ──────────── 1150/1150 2.8it/s 6:46
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-95)     Pose(P          R      mAP50  mAP50-95): 100% ──────────── 246/246 3.0it/s 1:23
                   all       7847       7847      0.947      0.901      0.972      0.736      0.878      0.696      0.746      0.519

      Epoch    GPU_mem   box_loss  pose_loss  kobj_loss   cls_loss    l1_loss   rle_loss  Instances       Size
```



Basic Prototype



The Process and Methodology



Decision making

Selected Pakistani Sign Language

Python As main language

Ultralytics as Deep learning library



Dataset Feeding

Trained the model on 20,000 Images of hands to track hands

100 Epoche on Colab and Kaggle (Yolo26n-pose)



Basic Program building

Python program to detect hands

Tested in different lighting



Pakistani Sign Lang Training

Program to feed signs to the model



Development

API Endpoints built

Portable Model

Built real App with our own language and culture

Project Preparation

To contribute to the deaf and disabled community of Pakistan is our ideology and objective, We believe technology should help the special Individuals around us

We have discovered there are very few sign language translators and ML models, and specifically no one has even tried for Pakistani Sign Language which translates to Urdu and local languages

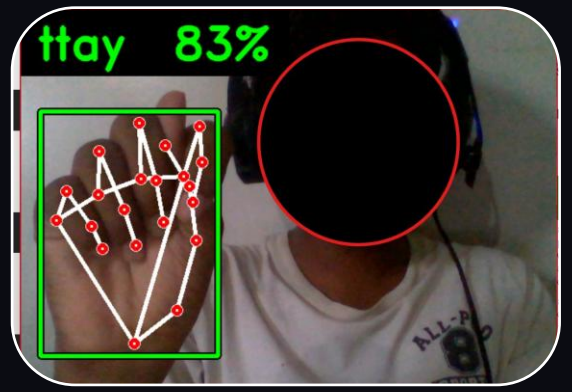
Our Project combines an Open source Machine learning model YOLO combined with media pipe holistic and Ultralytics which we fed a large dataset (~20,000 Images) of hands to detect hands , every step included careful testing after which we moved towards training it for signs so we developed a program to feed signs which now enables us to introduce sync as our own PSL-to-Speech translator while also open sourcing our API and Model for community of disabled individuals

The Initial risks and difficulties was to find a dedicated high-end Hardware to train the model, which we luckily found on colab and Kaggle and trained it on the large dataset for a whole day after a day of failure

Final Model & Demonstration



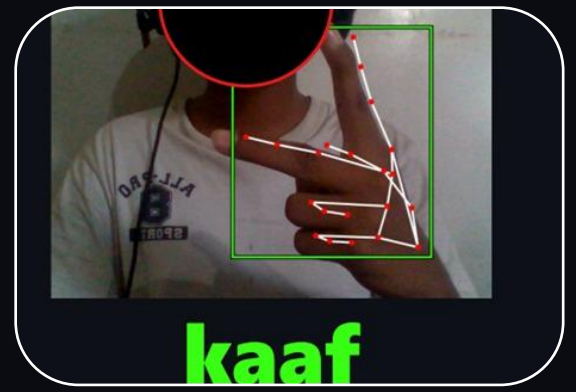
Words detection



In-App detection



Web based Detection



Letters detection

Project Evaluation

Successful Objectives Met

- Contribution to Disabled community
- Low-cost translation tool
- First Pakistani Sign Language translator
- Support 5+ local languages
- Communication made easier
- Given away to developer's community for further innovation

Results and Future Scope

Results:

Model working:

Fully Working!

Translates to:

5+ Languages

Including urdu , sindhi and memoni

Sign Languages covered:

Pakistani Only

Cost

Almost Free

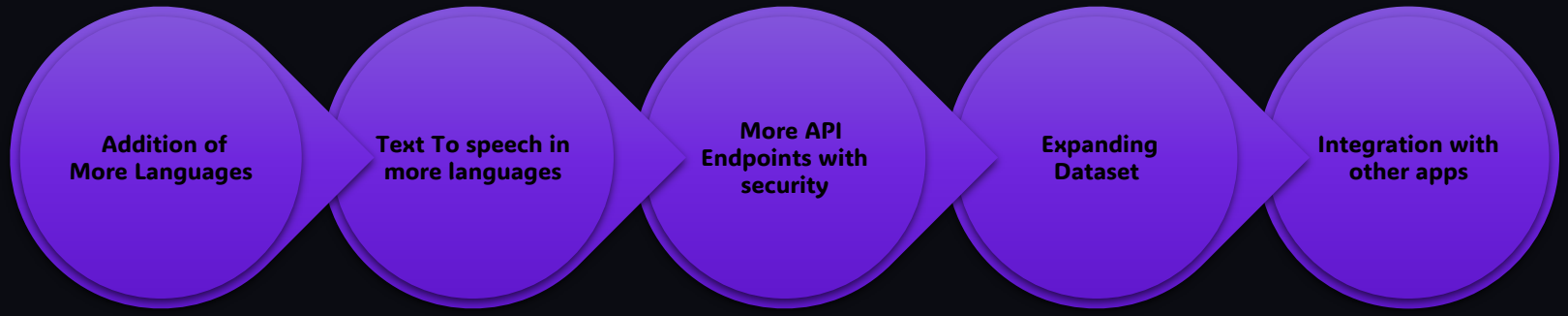
#1

Pakistani Signs Language translator

Dataset

1000+ Images and videos of signs

Future Scope:



Social Impact & Conclusion

With Sync , we can help disabled individuals of the society communicate more efficiently as well as specifically people of Pakistan who have a distinguished sign language which requires special attention while also translating to local languages like Urdu , Pashto , sindhi and still expanding because our culture should be our first. As Pakistan has high number of uneducated, we plan to fine-tune sync into free applications on all platforms for sign-to-speech while also developing further accuracy and speed of model.

To conclude , we have met most of our objectives, but the never-ending race of innovation and curiosity intrigues us to build and scale this further.

Thank You!